

COHORT RETENTION BASICS

Data Analytics Track – Python & SQL

1. Project Overview

Objective

The objective of this project is to build a **cohort retention table by signup month** and understand how customer activity changes over time.

Customers are grouped into cohorts based on the month in which they signed up. Their activity is then tracked across subsequent months to calculate the percentage of customers who remain active.

Tools & Technologies

- Python
 - Pandas
 - NumPy
 - SQL
 - Matplotlib
 - Seaborn
 - CSV Dataset
-

2. Dataset Description

The project uses a customer activity dataset containing information about customer signup dates and subsequent activity.

Important Columns

Column	Description
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CustomerID	Unique customer identifier
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SignupDate	Date on which the customer joined
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EventDate	Date of customer activity
-----------	---------------------------

Quantity	Quantity associated with the transaction
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UnitPrice	Price per unit
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Revenue	Quantity × UnitPrice
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The dataset is used to identify customer cohorts and measure their activity over time.

3. What is Cohort Analysis?

Cohort analysis is a method of grouping customers based on a common characteristic or event.

In this project, customers are grouped according to their **signup month**.

For example:

- January 2009 signup → January 2009 cohort
- February 2009 signup → February 2009 cohort
- March 2009 signup → March 2009 cohort

The activity of each cohort is then tracked month by month.

4. Retention Calculation

Customer retention is calculated using the following formula:

Retention % = (Active Unique Customers in a Month / Month 0 Cohort Size) × 100

Example

If a cohort contains 100 customers in Month 0 and 40 of those customers are active in Month 1:

Month 1 Retention = (40 / 100) × 100 = 40%

Month 0 is considered 100% because all customers in the cohort are active at the time they join.

5. Methodology

The following steps were performed:

1. Loaded the customer activity dataset using Pandas.
2. Converted signup and activity dates into monthly periods.
3. Created a signup-month cohort for every customer.
4. Calculated the number of months between signup and activity.
5. Removed duplicate customer activity records for accurate unique-customer counting.
6. Calculated the number of active customers for every cohort and month.
7. Calculated the cohort retention percentage.
8. Created a retention matrix.
9. Visualized the retention matrix using a heatmap.
10. Interpreted the results from a business perspective.

6. Python Implementation

The Python implementation uses Pandas to perform cohort analysis.

Main Logic

```
df["cohort_month"] = df["SignupDate"].dt.to_period("M").dt.to_timestamp()
```

```
df["activity_month"] = df["EventDate"].dt.to_period("M").dt.to_timestamp()
```

```
df["cohort_index"] = (  
    (df["activity_month"].dt.year - df["cohort_month"].dt.year) * 12  
    + (df["activity_month"].dt.month - df["cohort_month"].dt.month)  
)
```

Unique active customers are then calculated for every cohort and month.

The retention percentage is calculated as:

```
retention_pct = (  
    active_customers / cohort_size  
) * 100
```

7. SQL Implementation

SQL can also be used to perform cohort analysis.

The SQL workflow:

- Groups customers by signup month
- Calculates activity month
- Determines months since signup
- Counts unique active customers
- Calculates retention percentage

SQL Logic

```
SELECT  
    cohort_month,  
    cohort_index,
```

```

COUNT(DISTINCT CustomerID) AS active_customers,
cohort_size,
ROUND(
    100.0 * COUNT(DISTINCT CustomerID) / cohort_size,
    1
) AS retention_pct
FROM cohort_activity
GROUP BY cohort_month, cohort_index, cohort_size
ORDER BY cohort_month, cohort_index;

```

8. Cohort Retention Table

The resulting cohort table contains:

- Signup Cohort
- Month 0
- Month 1
- Month 2
- Month 3
- Month 4
- Month 5
- Subsequent months where data is available

Each cell represents the percentage of the original cohort that remained active during that month.

Interpretation

Higher percentages indicate stronger customer retention.

Lower percentages indicate that fewer customers from the original cohort remained active.

Blank cells represent months for which sufficient activity data is not available.

9. Cohort Retention Heatmap

The retention table is visualized using a heatmap.

The heatmap makes it easier to identify:

- High-retention cohorts

- Low-retention cohorts
- Customer drop-off over time
- Differences between signup cohorts
- Changes in retention across months

The heatmap provides a visual representation of customer engagement and retention.

10. Key Insights

Insight 1 – Month 0 Retention

Month 0 retention is **100% by definition**, because the cohort consists of customers who signed up during that month.

Insight 2 – Month 1 Retention

Month 1 is an important retention checkpoint because it indicates how many newly acquired customers returned after their initial signup.

Insight 3 – Long-Term Retention

Retention generally decreases as the number of months since signup increases. This indicates customer drop-off over time.

Insight 4 – Cohort Comparison

Different signup cohorts can show different retention patterns. Cohort analysis makes these differences easier to identify than looking only at overall customer counts.

Insight 5 – Business Opportunities

Large drops in retention can be investigated using:

- Customer onboarding improvements
 - Personalized offers
 - Email or notification reminders
 - Loyalty programs
 - Product improvements
 - Customer engagement campaigns
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11. Business Applications

Cohort retention analysis can be used by businesses to:

- Measure customer loyalty
- Understand customer lifecycle

- Evaluate acquisition quality
- Identify customer drop-off periods
- Compare different customer groups
- Improve marketing campaigns
- Improve customer onboarding
- Support customer lifetime value analysis

12. Project Deliverables

The project contains the following files:

File	Description
customer_events.csv	Customer activity dataset
cohort_retention_analysis.py	Complete Python analysis
cohort_retention.sql	SQL cohort-retention query
cohort_retention_table.csv	Retention percentage table
cohort_active_customers.csv	Active customer count table
cohort_retention_heatmap.png	Cohort retention visualization
insights.md	Business insights
README.md	Project documentation
requirements.txt	Requ